

List of Publications

Wasiur R. KhudaBukhsh

1 Preprints/Submitted

- [1] Kushankur Dutta, Olga Izyumtseva, Wasiur R. KhudaBukhsh, and Grzegorz A. Rempała. Coloured epidemic models: functional law of large numbers and propagation of chaos. 2026.
- [2] Olga Izyumtseva, Wasiur R. KhudaBukhsh, M. Gabriela M. Gomes, and Grzegorz A. Rempała. From individual-based stochastic epidemics to heterogeneous SIR equations. 2026. Submitted.
- [3] Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and Wasiur R. KhudaBukhsh. Shrinkage priors for Bayesian substitute confounders. 2026.
- [4] Arnab Ganguly and Wasiur R. KhudaBukhsh. Fluctuation analysis of the standard QSSA for a stochastic multistage Michaelis–Menten model. 2026. Submitted.
- [5] Olga Izyumtseva and Wasiur R. KhudaBukhsh. Self-intersection local times for Volterra Gaussian processes in stochastic flows with interaction. 2026. Submitted.
- [6] J. D. Harborne, Wasiur R. KhudaBukhsh, and John R. King. Deterministic and stochastic asymptotics for a multiscale epidemic model. 2026. Revision requested.
- [7] Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and Wasiur R. KhudaBukhsh. Kernel-based estimators for functional causal effects. 2026. Under review.
- [8] Wasiur R. KhudaBukhsh and Yangrui Xiang. Mixing time for a noisy SIS model on graphs. 2026. Submitted.
- [9] Karim S. Elsayed, Olga Izyumtseva, Wasiur R. KhudaBukhsh, and Amr Rizk. Stochastic analysis of entanglement-assisted quantum communication channels. 2025. Revision requested.
- [10] Olga Izyumtseva and Wasiur R. KhudaBukhsh. Local times of self-intersection and sample path properties of Volterra Gaussian processes. 2025. Revision requested.

2 Peer-reviewed journal publications

- [11] Arnab Ganguly and Wasiur R. KhudaBukhsh. Statistical inference for a multiscale stochastic model of enzyme kinetics via propagation of chaos. *Stochastic Processes and their Applications*, 2027. Accepted for publication.
- [12] Arnab Ganguly and Wasiur R. KhudaBukhsh. Asymptotic analysis of the total quasi-steady state approximation for the Michaelis–Menten enzyme kinetic reactions. *Journal of Mathematical Analysis and Applications*, 561(1):130551, 2026.
- [13] Yi Fu, Hye-Won Kang, Wasiur R. KhudaBukhsh, Lea Popovic, Grzegorz A. Rempała, and Ruth J. Williams. Fragility in a Togashi–Kaneko stochastic model with mutations. *SIAM Journal on Life Sciences*, 1(2):202–228, 2026.
- [14] Riccardo Corradin, Luca Danese, Wasiur R. KhudaBukhsh, and Andrea Ongaro. Model-based clustering of time-dependent observations with common structural changes. *Statistics and Computing*, 36(7), 2026.
- [15] Alexander E. Zarebski, Nefel Tellioglu, Jessica E. Stockdale, Julie A. Spencer, Wasiur R. KhudaBukhsh, Joel C. Miller, and Cameron Zachreson. Including frameworks of public health ethics in computational modelling of infectious disease interventions. *Interface Focus*, 15, 2025.
- [16] Yushuf Sharker, Zaynab Diallo, Wasiur R. KhudaBukhsh, and Eben Kenah. Pair-wise accelerated failure time regression models for infectious disease transmission in close contact groups with external sources of infection. *Statistics in Medicine*, 2024.
- [17] Wasiur R. KhudaBukhsh and Grzegorz A. Rempała. How to *correctly* fit an SIR model to data from an SEIR model? *Mathematical Biosciences*, 2024.
- [18] Matthew Wascher, Patrick Schnell, Wasiur R. KhudaBukhsh, Mikkel B.M. Quam, Joseph Tien, and Grzegorz Rempała. Estimating disease transmission in a closed population under repeated testing. *Journal of the Royal Statistical Society: Series C (JRSSC)*, 2024.
- [19] István Z. Kiss, Luc Berthouze, and Wasiur R. KhudaBukhsh. Towards inferring network properties from epidemic data. *Bulletin of Mathematical Biology*, 2024.
- [20] Wasiur R. KhudaBukhsh, Sat Kartar Khalsa, Eben Kenah, Grzegorz Rempała, and Joseph Tien. COVID-19 dynamics in an Ohio prison. *Frontiers in Public Health*, 2023.
- [21] Wasiur R. KhudaBukhsh, Caleb Deen Bastian, Matthew Wascher, Colin Klaus, Saumya Yashmohini Sahai, Mark H. Weir, Eben Kenah, Elisabeth Root, Joseph H. Tien, and Grzegorz A. Rempała. Projecting COVID-19 cases and hospital burden in Ohio. *Journal of Theoretical Biology*, 561:111404, 2023.

- [22] Colin Klaus, Matthew Wascher, Wasiur R. KhudaBukhsh, and Grzegorz Rempała. Likelihood-free dynamical survival analysis applied to the COVID-19 epidemic in Ohio. *Mathematical Biosciences and Engineering*, 20, 2023.
- [23] Kai Cui, Wasiur R. KhudaBukhsh, and Heinz Koepl. Hypergraphon mean-field games. *Chaos*, 2022.
- [24] Wasiur R. KhudaBukhsh, Casper Woroszylo, Grzegorz Rempała, and Heinz Koepl. A functional central limit theorem for SI processes on configuration model graphs. *Advances in Applied Probability*, 2022.
- [25] Colin Klaus, Matthew Wascher, Wasiur R. KhudaBukhsh, Joseph H. Tien, Grzegorz A. Rempała, and Eben Kenah. Assortative mixing among vaccination groups and biased estimation of reproduction numbers. *The Lancet Infectious Diseases*, 22:P579–581, 2022.
- [26] Francesco Di Lauro*, Wasiur R. KhudaBukhsh*, István Z. Kiss, Eben Kenah, Max Jensen, and Grzegorz Rempała. Dynamic survival analysis for non-Markovian epidemic models. *Journal of the Royal Society Interface*, 2022. *Joint first authors.
- [27] Kai Cui, Wasiur R. KhudaBukhsh, and Heinz Koepl. Motif-based mean-field approximation of interacting particles on clustered networks. *Physical Review E*, 105, 2022.
- [28] Harley Vossler, Pierre Akilimali, Yuhan Pan, Wasiur R. KhudaBukhsh, Eben Kenah, and Grzegorz A. Rempała. Analysis of individual-level epidemic data: study of 2018–2020 Ebola outbreak in Democratic Republic of the Congo. *Scientific Reports*, 12, 2022.
- [29] Ido Somekh*, Wasiur R. KhudaBukhsh*, Elisabeth Dowling Root*, Greg Rempała, Eric Simoes, and Eli Somekh. Quantifying the population-level effect of COVID-19 mass vaccination campaign in Israel: a modeling study. *Open Forum Infectious Diseases*, 2022. *Joint first authors.
- [30] Wasiur R. KhudaBukhsh*, Hye-Won Kang, Eben Kenah, and Grzegorz Rempała. Incorporating age and delay into models for biophysical systems. *Physical Biology*, 18(1), 2021. Invited paper.
- [31] Wasiur R. KhudaBukhsh, Boseung Choi, Eben Kenah, and Grzegorz Rempała. Survival dynamical systems: individual-level survival analysis from population-level epidemic models. *Journal of the Royal Society Interface Focus*, 10(1), 2020.
- [32] Wasiur R. KhudaBukhsh, Arnab Auddy, Yann Disser, and Heinz Koepl. Approximate lumpability for Markovian agent-based models using local symmetries. *Journal of Applied Probability*, 56, 2019.

- [33] Hye-Won Kang*, Wasiur R. KhudaBukhsh*, Heinz Koepl, and Grzegorz Rempała. Quasi-steady-state approximations derived from a stochastic enzyme kinetics. *Bulletin of Mathematical Biology*, 81(5):1303–1336, 2019. *Joint first authors.
- [34] Saumya Yashmohini Sahai, Saket Gurukar, Wasiur R. KhudaBukhsh, Srinivasan Parthasarathy, and Grzegorz A. Rempała. A machine learning model for nowcasting epidemic incidence. *Mathematical Biosciences*, 2021.
- [35] Wasiur R. KhudaBukhsh, Sounak Kar, Bastian Alt, Amr Rizk, and Heinz Koepl. Generalized cost-based job scheduling in very large cluster systems. *IEEE Transactions on Parallel and Distributed Systems*, 31(11):2594–2604, 2020.
- [36] Boseung Choi, Sydney Busch, Dieudonné Kazadi, Benoit Ilunga, Emile Okitolonda, Yi Dai, Robert Lumpkin, Omar Saucedo, Wasiur R. KhudaBukhsh, Joseph Tien, Marcel Yotebieng, Eben Kenah, and Grzegorz A. Rempała. Modeling outbreak data: analysis of a 2012 Ebola virus disease epidemic in DRC. *BIOMATH*, 8(2), 2019.
- [37] Wasiur R. KhudaBukhsh, Amr Rizk, Sounak Kar, and Heinz Koepl. Provisioning and performance evaluation of parallel systems with output synchronization. *ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)*, 4(1), 2019.
- [38] Bastian Alt, Markus Weckesser, Christian Becker, Matthias Hollick, Sounak Kar, Anja Klein, Robin Klose, Roland Kluge, Heinz Koepl, Boris Koldehofe, Wasiur R. KhudaBukhsh, Mahdi Mousavi, Martin Pfannemueller, Amr Rizk, Andy Schuerr, and Ralf Steinmetz. Transitions: a protocol-independent view of the future internet. *Proceedings of the IEEE*, 107(4):835–846, 2019.

3 Book chapters

- [39] Olga Izyumtseva, Wasiur R. KhudaBukhsh, and Grzegorz A. Rempała. Functional law of large numbers for an epidemic model with random effects. *Handbook of Statistics*. Elsevier, 2024.
- [40] Grzegorz A. Rempała and Wasiur R. KhudaBukhsh. Dynamical survival analysis for epidemic modeling. In *Handbook of Visual, Experimental and Computational Mathematics*, pages 1–17. Springer International Publishing, 2023.

4 Peer-reviewed conference proceedings

- [41] Karim Elsayed, Wasiur R. KhudaBukhsh, and Amr Rizk. On the trade-off between fidelity and latency for the quantum link layer with few memories and entanglement purification. In *Proceedings of the International Conference on Quantum Communications, Networking, and Computing (QCNC 2024)*, 2024. Best Paper Award.

- [42] Karim Elsayed, Wasiur R. KhudaBukhsh, and Amr Rizk. On the fidelity distribution of link-level entanglements under purification. In *Proceedings of the IEEE International Conference on Communication (ICC) 2024*, 2024.
- [43] Riccardo Corradin, Luca Danese, Wasiur R. KhudaBukhsh, and Andrea Ongaro. Model-based clustering of non-stationary time series with common historical change times. In *Statistical Learning, Sustainability and Impact*, 2023.
- [44] Wasiur R. KhudaBukhsh, Bastian Alt, Sounak Kar, Amr Rizk, and Heinz Koepl. Collaborative uploading in heterogeneous networks: optimal and adaptive strategies. In *IEEE International Conference on Computer Communications (INFOCOM)*, 2018. <20% acceptance rate. Best-in-Session Presentation Award.
- [45] Wasiur R. KhudaBukhsh, Amr Rizk, Alexander Frömmgen, and Heinz Koepl. Optimizing stochastic scheduling in fork-join queueing models: bounds and applications. In *IEEE International Conference on Computer Communications (INFOCOM)*, 2017. ~20% acceptance rate.
- [46] Adrian Šošić, Wasiur R. KhudaBukhsh, A. M. Zourbir, and Heinz Koepl. Inverse reinforcement learning in swarm systems. In *AAMAS Workshop on Transfer in Reinforcement Learning*, 2017. Available: <http://www.tirl.info/proceedings/2017/SosicEtal-Inverse.pdf>.
- [47] Adrian Šošić, Wasiur R. KhudaBukhsh, A. M. Zourbir, and Heinz Koepl. Inverse reinforcement learning in swarm systems. In *International Conference on Autonomous Agents & Multiagent Systems (AAMAS)*, 2017. ~26% acceptance rate. Best Paper Award finalist.
- [48] Wasiur R. KhudaBukhsh, Julius Rückert, Julian Wulfheide, David Hausheer, and Heinz Koepl. Analysing and leveraging client heterogeneity in swarming-based live streaming. In *IFIP Networking Conference (IFIP Networking) and Workshops*, 2016. ~26% acceptance rate.
- [49] Mahdi Mousavi, Hussein Al Shatri, Wasiur R. KhudaBukhsh, Heinz Koepl, and Anja Klein. Cross-layer QoE-based incentive mechanism for video streaming in multi-hop wireless networks. In *IEEE 86th Vehicular Technology Conference (VTC)*, 2017.

5 Thesis and technical notes

- [50] Wasiur R. KhudaBukhsh. *Model reductions for queueing and agent-based systems with applications in communication networks*. PhD thesis, Technische Universität, Darmstadt, 2018. Available at: <http://tuprints.ulb.tu-darmstadt.de/7588/>.

- [51] Mark Sinzger-D'Angelo, Heinz Koepl, and Wasiur R. KhudaBukhsh. Bounds on the spectral radius of real-valued non-negative kernels on measurable spaces. 2023. Preprint: <https://arxiv.org/abs/1808.00258>.